

SIDDHANT SANADHAYA

Applied Scientist | Generative AI & Multimodal Intelligence

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PROFESSIONAL SUMMARY

Accomplished **Applied Scientist & Software Development Engineer** with **6+ years of total technical and engineering experience** (including 5+ years specializing in Generative AI, Machine Learning, Computer Vision, and NLP). Proven track record in researching, designing, and deploying state-of-the-art **Agentic AI Orchestrator platforms**, multimodal workflows, and domain-specific AI solutions. Hands-on expert in custom multi-agent reasoning systems (LangGraph), automated ML/AutoML pipelines, and computer vision recognition models across cloud infrastructure (AWS, Azure, GCP). Experienced in driving applied research to production, conducting impactful Proofs of Concept (POCs), and delivering significant operational cost savings for enterprise clients.

CORE COMPETENCIES & TECHNICAL SKILLS

AI & Machine Learning:	Generative AI, Agentic AI Orchestration, Multimodal Learning (Text, Audio, Vision), LLMs, Transformers, Deep Learning, Recommendation Systems, Semantic Understanding.
Frameworks & Libraries:	PyTorch, TensorFlow, LangGraph, Hugging Face, OpenCV, Scikit-Learn, PySpark (Large-scale Semantic Analysis), AutoML.
Computer Vision & NLP:	Deep Learning-based OCR, Object Detection, Medical Image Analysis, Automated Data Annotation, NLP Semantic Pipelines.
Software & Infrastructure:	Python, RESTful APIs (FastAPI, Flask, Django), AWS (Lambda, S3, EC2), Azure, GCP, SQL/NoSQL, Git, CI/CD, Linux.
Applied R&D & Business:	Enterprise POC Development, Algorithmic Optimization & Cost Reduction, Pre-Sales/Post-Sales Technical Support, Multi-agent Architecture Design.

PROFESSIONAL EXPERIENCE

Software Development Engineer — AI & Generative Systems StatusNeo	01/2023 – Present
- Architected and deployed enterprise-grade Agentic AI Orchestrator solutions for procurement applications using LangGraph, executing multi-step reasoning agents and tool-based automated workflows. - Designed robust multi-user, multi-session state architectures with per-user chat persistence, real-time context management, and custom memory management to deliver tailored human-machine interaction experiences. - Engineered scalable e-commerce Machine Learning platforms, including State-of-the-Art (SOTA) Recommendation Systems ("Suggested Stores", "Buy Again") driving customer engagement and high NPS scores. - Developed large-scale semantic data pipelines leveraging PySpark and Transformer models for deep text analytics across massive unstructured enterprise datasets. - Implemented end-to-end AutoML pipelines across diverse ML frameworks, optimizing existing model training workflows and substantially reducing compute and operational overhead.	
Python / Machine Learning Developer — Computer Vision & Cloud McKinley Rice Creativity	03/2022 – 01/2023
- Spearheaded deep learning and Computer Vision research to significantly enhance accuracy and speed in medical image analysis pipelines using PyTorch, OpenCV, and custom neural network architectures. - Architected high-throughput, low-latency RESTful microservices and API gateways using FastAPI, Flask, and Django for seamlessly serving ML model inferences to production clients.	

- Designed automated Serverless Computer Vision processing pipelines deployed on **AWS Lambda**, enabling real-time image analysis and scalable cloud performance.
- Collaborated closely with product development teams and stakeholders to convert client business requirements into high-impact Proof of Concepts (POCs) and technical solutions.

Project Engineer — Deep Learning & OCR Systems
Rawattech

08/2020 – 03/2022

- Led cross-functional engineering efforts to research, design, and implement an offline **Deep Learning-based Multilingual Handwritten OCR system**, independently developing the core mathematical parsing and character recognition models.
- Directed the development of an automated **'Roll Number Detection System'**, integrating custom computer vision algorithms that increased project efficiency by **25%**.
- Provided pre-sales and post-sales technical consulting to business development teams, identifying customer business bottlenecks and proposing targeted AI-driven solution architectures.

KEY INNOVATIONS & APPLIED R&D PROJECTS

Automatic Image Annotation Tool (Computer Vision & Data Mining)

Rawattech / StatusNeo

- Conceived and developed an automated computer vision annotation system for object detection training datasets, achieving **\$20,000 in operational cost savings** and reducing manual annotation workload by **40%**.
- Accelerated dataset preparation throughput from 300–400 manual images/day to **1,000 auto-annotated images/hour**, drastically speeding up ML model training cycles.

GenAI Enterprise Technical Writing Assistant

StatusNeo

- Built a Generative AI platform integrating LLMs with Jira, Confluence, and Google Docs APIs via automated CI/CD pipelines, automating complex technical documentation generation and reducing technical writer workload by **>50%**.

EDUCATION

Diploma in Computer Programming / Data Science — Indian Institute of Technology (IIT), Madras

2023 – Present

B.Tech in Computer Engineering — Rajasthan Technical University, Jaipur, India

07/2016 – 07/2020